

# Excel Basics

## 1. Explain the difference between Relative , Absolute and Mixed Cell Referencing.

### i. Relative Cell Reference

- In relative referencing, the cell reference in a formula changes automatically when the formula is copied or moved to another cell. It adjusts according to the relative position of rows and columns. Used when you want the formula to adjust automatically for each row or column, such as when calculating totals for multiple rows of data.

Example: If cell C1 contains the formula =A1 + B1, and it is copied to C2, the formula automatically changes to =A2 + B2.

### ii. Absolute Cell Reference

- In absolute referencing, the cell reference does not change when copied or moved. It remains fixed to a specific cell, regardless of where the formula is placed. An absolute reference is indicated by adding dollar signs (\$) before the column letter and row number.

Used when a formula must always refer to a constant value, such as a fixed tax rate, discount rate, or conversion factor stored in one specific cell.

Example: = \$A\$1 + B1 — if copied to another cell, it remains = \$A\$1 + B1.

### iii. Mixed Cell Reference

- In mixed referencing, either the row or the column is fixed, but not both. It combines features of relative and absolute referencing. Used when copying formulas across rows or columns where only one part of the reference (row or column) should remain constant.

Types of Mixed References:

- \$A1 → Column A is fixed, row changes when copied.
- A\$1 → Row 1 is fixed, column changes when copied.

Example: If = \$A1 is copied across columns, the column remains A, but the row changes according to the new position.

## 2. Write a formula to calculate the total Price of Car and Bicycle only.

-> Total price of car and bicycle is 510500.

Formula = SUM(F7,F8) (Taken from the Excel file that I have attached).

## 3. Using the data below, write a formula to calculate the average sales of items priced above 100 but less than 300.

-> The average sale of item price above 100 but less than 300 – 1350.

Formula = AVERAGEIFS(E10:E12,D10:D12,">100",D10:D12,"<300") (Taken from the Excel file that I have attached).

